MATH 250-01 (FALL 2025)
STATISTICAL METHODS I

Course Outline

Synchronized Online Component & Onsite Tests

*** *Classes will be held online in real-time, and tests will be conducted in person.* ***
Test dates: September 11, October 2, October 30, and November 20 & **Classroom:** ECTR 116

Instructor Information:
Name: Jin-Hong Park, PhD in Statistics
Office: 351 Robert Scott Small
Email: parkj@cofc.edu
Office Hours: 12-1 PM (TR); For additional assistance, you can reach out via email or schedule a personal Zoom meeting by appointment.

OAKS will serve as the platform for distributing the syllabus and class materials throughout the semester, with regular updates.

Course Meeting: Zoom Meeting at 1:40 PM – 2:55 PM on Tuesday and Thursday
* *It is mandatory for all students to have a computer with a web camera, microphone, and internet connection. Ensure your camera is on to allow accurate attendance tracking via Zoom. Additionally, keep your speaker muted unless you are speaking to minimize background noise.*

Course objectives: This course is designed to develop quantitative skills along with critical and interpretive judgment. It focuses on both methodology and applications in statistics. It provides students with advanced courses in statistics or methodological tools in their research disciplines, by approaching real-world problems and analyzing real data. Emphasis will be placed on the elements of sound statistical reasoning.

Course Prerequisite: MATH 116 with a C- or better or MATH 111 or MATH 120

Course Student Learning Outcomes: By the end of the course, students will be able to
1. summarize data by using methods of descriptive statistics,
2. choose appropriate inferential procedures and apply them to make inferences about populations,
3. understand the capabilities and limitations of statistical methods,
4. use appropriate technology, such as R, to perform various statistical procedures,
5. interpret results and conclude statistical analysis.

General Education Student Learning Outcomes: **Math/Logic (Problem Solving)**
In this course, we will provide evidence for assessment of Problem Solving. Students will apply appropriate methods to obtain conclusions. More specifically, students will be able to:
1. Clearly identify or define a problem.
2. Use a discipline-specific model to perform an analysis or procedure.
3. Draw a well-supported logical conclusion.

This outcome will be assessed in the small-scale projects, tests, and/or final term project.

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Statistical Software: Students will be taught to perform computations using the statistical software R, which is a renowned and free package widely used in statistical computing. It is required for students to have their own computers to install and utilize R throughout the semester.

Textbook or Reference: (*No other technologies except for R are required.)*
Neil A. Weiss, *Introductory Statistics* (10th Edition), Pearson Addison Wesley

Course Topics:
Descriptive Measures (Ch.3)
Probability Concepts (Ch.4)
Discrete Random Variables (Ch.5)
The Normal Distribution (Ch.6)
Sampling Distribution of the Sample Mean (Ch.7)
Confidence Intervals for One Population Mean (Ch.8)
Hypothesis Tests for One Population Mean (Ch.9)
Inferences for Two Population Means (Ch.10)
Inferences for Population Standard Deviations (Ch.11)
Inferences for Population Proportions (Ch.12)
Chi-Square Procedures (Ch. 13)
Regression and Correlation (Ch.14&15)
Analysis of Variance (Ch.16)

Grading Policy:
1. Homework: Problems provided as Problem Sets in the study guides are designed for individual study. While submission is not mandatory, they are crucial for understanding the course material and preparing for tests and the final term project.
2. **Small-Scale Projects (19%)**: These take-home assignments, including R practice, must be submitted on time. Late submissions will not be accepted unless an extension is granted by the instructor before the due date.
3. **Four Tests (76%):** The course includes four in-class tests, **each** constituting **19%** of the total course grade. **Make-up tests are not available**, so please carefully check test dates in advance.
4. **Final Term Project (5%):** The final term project constitutes 5% of the total grade.

All work submitted on this course must be your own, completed in accordance with the College's Honor Code. Unauthorized collaboration or use of ChatGPT or other GAI tools is prohibited.

To prevent any violation of academic honesty, answers and R commands not derived from our class materials will not be eligible for credit. If any instances of copying or sharing are discovered, a failing grade (F) will be assigned.

Project Policy: A student will receive an F in this course if they fail to submit more than two small-scale projects. Answers lacking a detailed procedure will not be eligible for credit. Similarly, R outputs without accompanying interpretations will not be accepted for credit. Work should be presented neatly and in an organized fashion.

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Grades:
A: 90 or above; A-: 87-89; B+:83-86; B: 80-82; B-: 77-79; C+:73-76; C: 70-72; C-: 67-69; D+: 63-66; D: 60-62; D-:57-59; F: 56 or below

Attendance Policy:
1. All students are expected to fully participate in every class.
2. Any student absent from five classes will receive an F for the course.

* *You are not required to submit any excuse documentation. The camera must be active during all class sessions. Your face and desk should be visible to ensure Zoom records your attendance. Failure to display your face and desk, or having the camera off, will result in an absence. Being more than 10 minutes late or leaving early will also be considered an absence.*
* *It is the student's responsibility to compensate for any missed classes due to absences.*

Midterm Policy: The instructor advises that students should consider withdrawing from the course if their midterm grade falls below 60%.

Important Dates:
August 25: Last day of drop/add
September 11: Test 1 - Classroom: ECTR 116
October 2: Test 2 (covers the first half of this course) - Classroom: ECTR 116
October 14: Fall Break (no class)
October 24: Last day for students to withdraw with a "W"
October 30: Test 3 - Classroom: ECTR 116
November 20: Test 4 (covers the second half of this course) - Classroom: ECTR 116
November 25: Final term project instructions provided
December 8: (3PM) Final Term Project Due

Recording of Classes (via ZOOM): Class sessions will be recorded via both voice and video recording. By attending and remaining in this class, the student consents to being recorded. Recorded class sessions are for instructional use only and may not be shared with anyone who is not enrolled in the class. The main goal of the recording is that students make up the classes for all absences.

College Honor Code: Any violation of the College's Honor Code will be reported to the Honor Board. For more details, see https://charleston.edu/student-handbook/handbook/honor-code.php and the Student Handbook at https://charleston.edu/student-handbook/index.php/

Accommodations for Students with Disabilities: If there is a student in this class who has a documented disability and has been approved to receive accommodations through the Center for Disability Services/SNAP (Students Needing Access Parity), please discuss this with me during my office hours. The exams require proctoring due to the use of R software, which I must personally oversee. This makes it impossible to arrange an alternative date or format for the exams.

Attendance Verification: Only students officially registered (graded or auditing) for this course may attend class. During the week following the drop/add deadline, the professor will verify student enrollment in this course. Any student appearing on the class roll but determined not to have attended the class even once will be removed.

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Important Notes:

1. I will frequently use email and OAKS to communicate important schedules. You must regularly check your email and OAKS. For email communication, please include *"Math 250"* in the subject line.
2. The course syllabus serves as a general guide; however, the instructor may need to make announced changes as necessary.
3. According to college policy, *Test 4 and the final term project* will not be returned to you.